

Chapter – 2

Helios Watch Store E-Commerce Database

2.0 Introduction

The entities required for the Helios Watch Store E-Commerce Database are identified based on business requirements. The attributes of each entity, along with their primary keys, foreign keys, unique keys, and attribute types, are determined to ensure proper data organization. The relationships between the entities are also identified to represent the interactions within the system and to establish a well-structured relational database design. In addition, the cardinality and participation constraints of each relationship are analyzed to define how entities are associated with one another. This analysis provides a clear understanding of the data requirements and serves as the foundation for designing the Entity Relationship (ER) diagram, relational schema, and subsequent database implementation while ensuring data integrity, consistency, and minimal redundancy.

2.1 Entity: Customer

Attribute	Key Type
Customer_ID	Primary Key
First_Name	Not Null
Last_Name	-
Email	Unique Key
Phone	-
Date_of_Birth	-
Age	Check
Address	Not Null

Entity: Brand

Attribute	Key Type
Brand_ID	Primary Key
Brand_Name	Unique Key
Country	-

Entity: Category

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

Entity: Product

Attribute	Key Type
Product_ID	Primary Key
Product_Name	-
Brand_ID	Foreign Key
Category_ID	Foreign Key
Price	-
Color	-
Strap_Material	-
Dial_Color	-
Water_Resistance	-
Stock	Not Null

Entity: Cart

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Grand_Total	-
Total_Items	-

Entity: Order

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Order_Status	-

Total_Amount	-
--------------	---

Entity: Order_Item

Attribute	Key Type
OrderItem_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-
Unit_Price	-

Entity: Payment

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Status	-
Payment_Date	-
Amount	-

Entity: Review

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	-
Review_Text	-
Review_Date	-

Entity: Shipment

Attribute	Key Type
Shipment_ID	Primary Key

Order_ID	Foreign Key
Delivery_Address	-
Shipment_Date	-
Delivery_Date	-
Shipment_Status	-

Entity: Warranty

Attribute	Key Type
Warranty_ID	Primary Key
Product_ID	Foreign Key
Warranty_Period	-
Start_Date	-
End_Date	-

2.2 Entity Relationships

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Partial / Total
Customer	Places	Order	One-to-Many	Partial / Total
Customer	Writes	Review	One-to-Many	Partial / Total
Brand	Manufactures	Product	One-to-Many	Partial / Total
Category	Categorizes	Product	One-to-Many	Partial / Total
Cart	Contains	Cart_Item	One-to-Many	Total / Total
Cart_Item	Refers To	Product	Many-to-One	Total / Partial
Order	Includes	Order_Item	One-to-Many	Total / Total
Order_Item	Refers To	Product	Many-to-One	Total / Partial
Order	Has	Payment	One-to-One	Total / Total
Order	Has	Shipment	One-to-One	Total / Total
Product	Receives	Review	One-to-Many	Partial / Total
Product	Has	Warranty	One-to-One	Total / Total